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Unsupervised Learning of Gaussian Mixture Model with Application to Image Segmentation*

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Abstract — Density estimation via Gaussian mixture modeling has been successfully applied to image segmentation, speech processing and other fields relevant to clustering analysis and Probability density function (PDF) modeling. Finite Gaussian mixture model is usually used in practice and the selection of number of mixture components is a significant problem in its application. For example, in image segmentation, it is the donation of the number of segmentation regions. The determination of the optimal model order therefore is a problem that achieves widely attention. This paper proposes a degenerating model algorithm that could simultaneously select the optimal number of mixture components and estimate the parameters for Gaussian mixture model. Unlike traditional model order selection method, it does not need to select the optimal number of components from a set of candidate models. Based on the investigation on the property of the elliptically contoured distributions of generalized multivariate analysis, it select the correct model order in a different way that needs less operation times and less sensitive to the initial value of EM. The experimental results show the effectiveness of the algorithm.

Key words — Gaussian mixture model, Model order, Degenerating model, Elliptically contoured distributions, Image segmentation.

I. Introduction

In computer vision and image processing the segmentation of an image is the division of a given (digital) image into contiguous regions. Image segmentation is an initial and vital step in a series of processes aimed at overall image understanding. Applications of image segmentation include identifying objects in a scene for object-based measurements such as size and shape, recognizing objects in a moving scene for object-based video compression (MPEG4), or objects which are at different distances from a sensor for a mobile robot[1].

In current computer vision algorithms the similarity of image parts is usually defined in terms of color and texture. The goal to automatically segment images into semantically meaningful parts is very difficult to achieve. Many methods have been proposed to deal with image segmentation. Although these methods may seem quite different, most of them belong to two categories, one based on clustering pixels and the other edge detection. To analyze the features inside the image, model based segmentation algorithms will be more efficient compared to non-parametric methods. As a powerful modeling tool of the former category, much work has been reported regarding image segmentation based on finite Gaussian mixture model (GMM) using EM algorithm in recent years. Mixture models could achieve stable results when using to image segmentation. The effectiveness of mixture model approach rests with the mixture density to model any unknown Probability density function (PDF)[2,3].

It has been widely acknowledged that there are two major difficulties when applying mixture models with EM. First, EM is not a global convergence algorithm which will only find a local maximum[4]. This could be avoided by starting EM with a better initialization approach such as k-means. The second important problem is model selection, i.e. how many mixture components are proper. Aiming at deciding the model order, many model selection criterions have been proposed, such as Bayes criterion, Mini description length (MDL), Bayesian information criterion (BIC), Akaike information criterion (AIC), Minimal message length (MML), *etc.* The idea of these methods is to add some penalty function to the negative log-likelihood expression of the optimal mixture model to a given data set and then minimize the expression with the penalty function. Thus the main work of such type methods focuses on constructing a better penalty. Figueiredo and Jain also introduced a component density annihilation mechanism with an extended EM to decide the model order[3,6,19,20]. All these methods need to choose a range of candidate models and the best one is then selected from these candidates. The parameters estimation of the mixture model and model order selection must be separated is another drawback of these methods. Many experiments have also illustrated that some of these

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methods, for example, AIC, tend to over-fit the model[19−22].

This paper mainly focuses on the second problem. A new algorithm is proposed that could automatically select the optimal number of components and estimate the mixture parameters.

II. Image Segmentation Using Gaussian Mixture Model

1. Finite Gaussian mixture model and EM algorithm

A finite d-dimensional Gaussian mixture model is a weighted sum of M Gaussian densities, as given by the following equation

$$p(x|\Theta) = \sum_{i=1}^{M} \alpha_i p_i(x|\theta_i) \tag{1}$$

$$p_i(x) = \frac{1}{(2\pi)^{d/2}|\boldsymbol{\Sigma}_i|^{1/2}} \exp\left\{ -\frac{1}{2}(x-\mu_i)^T \boldsymbol{\Sigma}_i^{-1}(x-\mu_i)\right\}$$

The parameters of the GMM are $\Theta = (\alpha_1, \alpha_2 \cdots \alpha_M, \theta_1, \theta_2 \cdots \theta_M)$ constrained by

$$\sum_{i=1}^{M} \alpha_i = 1 \tag{2}$$

Each p_i in the mixture density is parameterized by θ_i which indicates the mean vector $\boldsymbol{u}_i$ and covariance matrix $\boldsymbol{\Sigma}_i$ in a Gaussian distribution. Diagonal covariance matrix is often used in practice.

When GMM is used in image segmentation, at each pixel in an image, a d-dimensional feature vector $\boldsymbol{x}$ could be obtained and these feature vectors compose a sample set X. The image model assumes each pixel is produced by a density p_i associated with one of M image segment.

EM algorithm is commonly used to compute the Maximum likelihood (ML) parameters for the finite mixture model. EM is presented as an algorithm to perform ML parameter estimation with missing data. The EM algorithm works iteratively by alternately applying the two steps: the E-Step to compute the conditional expectation of the logarithm of the complete of the posteriori probability function and the M-Step to update the parameters. This process continues until some stopping criterion is met.

2. An SQP based EM algorithm for mixture model

For some given mixture component densities, for example, Gaussian density, it is possible to find an analytical expression for θ_i. If the component density has a complicated form, however, other methods might be sought for getting the solution θ_i.

Given an incomplete observed data set $X = \{\boldsymbol{x}_1, \boldsymbol{x}_2 \cdots \boldsymbol{x}_n\}$ where $\boldsymbol{x}_i$ is an N dimensional vector, the log likelihood expression for the mixture model Eq.(1) is

$$\begin{aligned}\log \prod_{i=1}^{n} p(x_i|\Theta) &= \sum_{i=1}^{n} \log p(x_i|\Theta) \\ &= \sum_{i=1}^{n} \log \sum_{j=1}^{M} (\alpha_j p_j(x_i|\theta_j))\end{aligned} \tag{3}$$

In E-Step of the EM, the posteriori probability could be obtained from Bayes law

$$P(i|x_j,\Theta) = \frac{\alpha_i p_i(x_j|\theta_i)}{p(x_j|\Theta)} = \frac{\alpha_i p_i(x_j|\theta_i)}{\sum_{i=1}^{M} \alpha_i p_i(x_j|\theta_i)} \tag{4}$$

In M-Step, the log-likelihood function could be written as:

$$Q(\Theta) = \sum_{j=1}^{N}\sum_{i=1}^{M} \log(\alpha_i p_i(x_j|\theta_i))P(i|x_j,\Theta) \tag{5}$$

$$\begin{aligned}Q(\Theta) =& \sum_{j=1}^{N}\sum_{i=1}^{M} \log(\alpha_i)P(i|x_j,\Theta) \\ &+ \sum_{j=1}^{N}\sum_{i=1}^{M} \log(p_i(x_j|\vartheta_i))P(i|x_j,\Theta) \\ =& I_1 + I_2\end{aligned} \tag{6}$$

where N denotes the number of training feature vectors.

To maximize $I_1 + I_2$, I_1 and I_2 could be respectively maximized since the parameters in them are not related[5].

The mixture weight in I_1 could be obtained by finding the solution of Lagrange equation with the constraint $\sum_{i=1}^{M} = \alpha_i = 1$

$$\frac{\partial}{\partial \alpha_i}\left[\sum_{i=1}^{M}\sum_{l=1}^{N} \log(\alpha_i)P(i|x_l,\Theta) + \lambda\left(\sum_{i=1}^{M}\alpha_i - 1\right)\right] = 0,$$

$$\alpha_i = \frac{1}{N}\sum_{j=1}^{N} P(i|x_j,\Theta) \tag{7}$$

It is observed that $\max(I_2) = \min(-I_2)$ and this is a multivariable optimization problem where the independent variable y could be rewritten as:

$$y = (u_1, u_2 \cdots u_n, \sigma_1, \sigma_2 \cdots \sigma_n, \tau_1, \tau_2 \cdots \tau_h)'$$

where $u_1, u_2 \cdots u_n$ and $\sigma_1, \sigma_2 \cdots \sigma_n$ are the elements of the mean vector and the diagonal covariance matrix. $\tau_1, \tau_2 \cdots \tau_h$ denote other relevant parameters in p_i. Sequential quadratic programming (SQP) could be applied to solve the problem. SQP is the most efficient and accurate method developed in recent twenty years for medium-scale nonlinear constrained optimization problem. It has the characteristics of high convergence speed (superlinear convergence)[7,10,11].

A positive definite approximation is made of the updated Hessian of Lagrange function using BFGS method at each iteration (the formula of Broyden, Fletcher, Goldfarb, and Shanno). This updated Hessian is then used to generate a Sub-QP problem and its solution is used to represent a search direction for a line search procedure. Further details and discussion of SQP could be found in Refs.[8-10].

3. Image segmentation using Gaussian mixture model

The EM algorithm is designed to deal with the type of missing data problem but it frames it slightly differently in image segmentation. It regards the segment as a missing data estimation problem. The incomplete data are just the measured pixel grey-levels or feature vectors.

The complete data are the measured grey levels or feature vectors plus a mapping function which indicates the labeling of each pixel. Given the complete data (pixels plus labels) the parameters of the mixture model can be easily worked out. Applying EM to image segmentation is to assume the image pixels or feature vectors follow a mixture component[12].

The expression of the labels is:

$$L_i = \max\left\{\frac{\alpha_i * \exp\left\{-\frac{1}{2}(x_j-u_i)\boldsymbol{\Sigma}^{-1}(x_j-u_i)^T\right\}}{|\boldsymbol{\Sigma}_i|^{1/2}}\right\},$$

$$i = 1, 2 \cdots M, \qquad j = 1, 2 \cdots N \tag{8}$$

In E-step, choose a number of segments and then initialize the parameters from small blocks of pixels and then compute the mixture weights.

Update the segments parameters in M-step until some convergence criterion is matched.

The complete procedure of image segmentation using GMM could be written as:

Initialize the Gaussian mixture model
Repeat
E-step:
Obtain an estimate of the labels based on the current parameter estimates
M-step:
Update the parameter estimates based on the current labelling
Until Convergence

III. Unsupervised Learning of Finite Mixture Model Using Degenerating Model Algorithm

1. Generalized multivariate analysis and Kotz Type distribution

The foundation and focus of traditional multivariate analysis is multivariate Gaussian distribution. Generalized multivariate analysis developed in recent twenty years extends multivariate Gaussian distribution to a more general form. The fundamentals of generalized multivariate analysis are elliptically contoured distributions, which were first put forward by Kelker in 1970 and widely discussed by K.T. Fang in 1990. As direct generalizations of the multivariate Gaussian distribution, which has dominated statistical theory and applications for almost a century, elliptically contoured distributions include many useful distributions such as well known Kotz distribution that will be introduced later. The contoured lines of such distributions have similar elliptical shapes like Gaussian distribution. Elliptically contoured distribution is a generalization of multivariate Gaussian distribution and therefore shares many of its tractable properties. This generalization for multivariate Gaussian distribution provides an attractive tool for multivariate analysis[13,14].

Kotz type distribution

Definition

An elliptical vector $\boldsymbol{x}$ belongs to the multivariate Kotz-type distribution, if its density function could be written as:

$$f(x) = \frac{C_n}{\sqrt{|\boldsymbol{\Sigma}|}}[(x-u)^T\boldsymbol{\Sigma}^{-1}(x-u)]^{K-1} \cdot \exp\{-r[(x-u)^T\boldsymbol{\Sigma}^{-1}(x-u)]^s\} \qquad (9)$$

with the normalizing constant.

$$c_n = s\pi^{-\frac{n}{2}} r^{\frac{2K+n-2}{2s}} \frac{\Gamma\left(\frac{n}{2}\right)}{\Gamma\left(\frac{2K+n-2}{2s}\right)}, \quad r,s>0, \quad 2K+n>2$$

The Kotz-type family includes several important distributions. Gaussian distribution is a member of this family having the parameter $K = 1$, $s = 1$, and $r = 1/2$. When $s = 1$ alone, it degenerates into the original Kotz distribution suggested by Kotz. When $K = 1$ alone, it is just an exponential distribution.

When $K < 1$, the distribution has a bell-shape density function similar to Gaussian distribution and tends to infinity at the mean. When $K > 1$, the density function has a local minimum at the mean and the distribution forms a space which looks like a crater. Inside the crater the density function approximates zero.

When $K > 1$, Kotz-type density has a useful property that, with K becoming larger, the space of the crater accordingly becomes broader. This property is helpful to decide the optimal number of components for GMM[13−15,18]. Figs.1 and 2 demonstrate the Kotz type density where $K = 2$ and $K = 20$ ($s = 1, r = 1/2$).

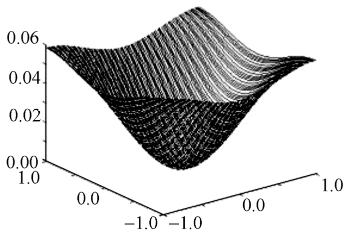

Fig. 1. Kotz-type $K = 2$

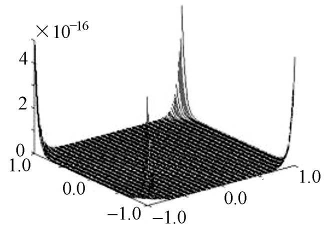

Fig. 2. Kotz-type $K = 20$

2. Analysis of the Kotz type mixture model where $K >= 1$

In essence, the Kotz type density include two sub-families, the bell-shape density (when $K <= 1$) and the crater-shape density (when $K > 1$). It has been investigated theoretically that if the data were from a multivariate Gaussian distribution, the Gaussian based ML estimator (*i.e.* Kotz type density when $K = 1$) perform better as expected than the Kotz type (when $K > 1$ or $K < 1$) ML estimator and vice versa[16]. Namely, the bell-shape density could produce a larger ML value when it describes a set of Gaussian type samples than a Kotz-type density (when $K > 1$). This illustrates that when a Gaussian type data set is described by a Kotz-type density ($K >= 1$), in order to maintain the ML value, the Kotz type density should degenerate into Gaussian density.

Furthermore suppose a data set that is described by an M components Kotz type mixture model. If the number of mixture components is smaller than optimal, the Kotz type component should degenerate into Gaussian component. If the number of mixture components is larger than optimal, the extra Kotz type components should extend its crater region to maintain the ML value of the mixture model.

Table 1. Training Kotz type mixture model with iris data set

Mixture number / Component number	2	3 (optimal)	4	5	6	7	8	9	10
Gaussian components	2	3	3	3	3(9times) 2(1times)	3	3(8times) 2(2times)	3	3
Kotz type components ($K > 1$)	0	0	1	2	3(9times) 4(1times)	4	5(8times) 6(2times)	6	7

To further explain that the Gaussian component can better fit a bell-shape cluster in a data set than the Kotz type component where $K > 1$, the Kotz type mixture model where $K > 1$ is used to fit iris data set that includes three clusters. The number of mixture components varies from 2 to 10. The SQP based EM run 10 times to train each Kotz type mixture model. The number of Gaussian and Kotz type components in a Kozt type mixture model is listed in Table 1. (**Note: the detail illustration of iris data set and experimental condition is list in Section IV**).

It could be seen from the table that when the number of mixture components is less than three, all the Kotz type mixture components degenerate into Gaussian distribution. The experiment result shows that in most cases, the Gaussian components could better fit a data cluster than a Kotz type component where $K > 1$. When the mixture number is 6 and 9, it can be noticed that sometimes the Gaussian components number is not equal to the actual class number of iris set. This is because some Kotz type components ($K > 1$) contribute a larger likelihood for the mixture model.

This experiment further confirm that the Kotz type components of a Kotz type mixture model would degenerate into Gaussian component $K > 1$ when the training data is sufficient as discussed above. This property could be used to decide the optimal number of components for the Gaussian mixture model.

3. Degenerating model

The phenomenon, whereby the Kotz type components degenerate into Gaussian components could be explained as another process. When the number of Gaussian components in a GMM is less than optimal number, another Kotz type component to mix with this GMM would degenerate into a Gaussian distribution. When the number of Gaussian components is equal to or larger than the optimal number, the parameter K in the mixed Kotz type density would become larger than 1 to make the space of the crater become broader, *i.e.* to maintain the ML value of the mixture model, the entire training feature vectors would fall into the crater and their probabilities approach zero. The Kotz type density where $s = 1$, $r = 1/2$ and $K \in [1, H]$ ($H > 1$) is named as degenerating model for GMM.

If the degenerating model degenerates into a Gaussian component with a very large covariance value when the number of Gaussian components of a GMM has been optimal, the degenerating model would lead to misjudgment. So the covariance value of degenerating model should be limited to an upper bound.

4. Defining the crater

In III.1 and III.2, the M clusters is assumed to have a Gaussian shape. Though this assumption is reasonable in most cases, sometimes a Kotz type component might better fit a data cluster than several Gaussian components (as the number of components is 6 or 9 in Table 1). So only the criterion $K > 1$ could not judge if the number of Gaussian components is optimal. The crater region of the degenerating model should be defined to identify those clusters which do not obey the distribution of degenerating model (when the number of components is less than the optimal number), though Table 1 shows the number of such components is very small.

The definition includes the following points. The estimate of K of the added Kotz type component is limited to $1 \cup [H_1, H_2]$. The value of the lower bound H_1 must make the space of the crater become broader enough to cover all the training feature vectors. The GMMs with different number of components are well trained in advance and their ML values converge to global maximum. The start value of K is bigger enough to make the space of the crater cover the feature space of the training data. The search direction of K in non-linear programming is from big to small. If its initial equals 1, the Kotz type may also degenerate into a Gaussian component with vey large covariance when GMM is over-fitted.

5. Complete degenerating algorithm

(1) Suppose a GMM with M Gaussian components as shown in Eq.(1)

(2) The GMM comprised M Gaussian components is well trained.

(3) Train the new mixture composed of the M-th order GMM and the degenerating model and store the operation result of parameter K

(4) If the value of stored K changes, stop;

Else if $K > 1$, $M = M - 1$ and turn to step (2);

Else if $K = 1$, $M = M + 1$ and turn to step (2)

IV. Experimental Results

Experiment 1

In order to illustrate the proposed degenerating model algorithm is adaptive to general clustering analysis, iris data set is used in the first experiment. The constraint of $1 \cup [H_1, H_2]$ is set as $1 \cup [10, 100]$.

Iris set is a classical data set including 150 samples from Fisher used for discriminant analysis. These samples belong to three classes and each class has 50 samples. The k-means algorithm is used in the initialization of EM.

As a most commonly used method for model selection, BIC criterion is used to compare the experimental results with the degenerating model algorithm. BIC criterion can be written as $-L(X, \Theta) + \dfrac{p}{2} \log N$ where L indicates the Log-Likelihood of the mixture, p indicates the number of free number of parameters in the mixture model and N is the dimension of the feature vectors. BIC starts by obtaining a set of candidate models where the model order ranging from $K_{\min}$ to $K_{\max}$. The goal is to find an optimal model from these candidate models which is assumed to contain the optimal one. In this experiment, the number of classes of the iris set is three, so the range of candidate model orders is from 2 to 10.

The degenerating model algorithm runs 50 times and each time it gives the correct selection 3.It can be seen from the experiment that when the number of mixture components is larger than 3, the value of parameter K in the degenerating model ranges from 7 to 15. It could be further calculated that all the probabilities of the training data are less than 10-3e when K is within this scope. This illustrates all the training data fall into the crater region of the degenerating model.

BIC also correctly selected the model order, but it could not simultaneously compute the parameters of GMM and select its model order. The degenerating model algorithm combines the parameters estimation and selection of model order into one procedure. Obviously the proposed algorithm has the advantage of high efficiency and convenient for use. Other methods such as AIC, BIC also need to give a scope of candidates' model for the given data set.

Experiment 2

To further study the performance of degenerating model algorithm, the second experiment considers the components of a GMM overlap, *i.e.*, parts of the mean points of Gaussian components are equal.

AIC and BIC are used in this experiment to compare their performance with the proposed algorithm.

First, a distribution that could be fitted by three Gaussian components is generated with 300 2-dimensional samples. Two of Gaussian components are located at the same mean point but have different covariance matrixes. The parameters of the mixture are shown as follows.

$$\alpha_1 = 0.2, \alpha_2 = 0.3, \alpha_3 = 0.5, u_1 = u_2 = (3,1)^T, u_3 = (2,4)^T$$
$$C_1 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, C_2 = \begin{pmatrix} 1.5 & 1 \\ 1 & 1.5 \end{pmatrix}, C_3 = \begin{pmatrix} 2.5 & -1 \\ -1 & 2.5 \end{pmatrix}$$

The algorithm runs 50 times and the proposed algorithm selects the correct model order three. BIC also selects the correct number every time. The operation times between BIC and the degenerating model algorithm is much different. Using the initialization method described in Section III, the degenerating model algorithm only need $2 \sim 3$ operation times including the initialization and operation of the algorithm. BIC, however, needs more iteration times (at least four times) since it needs to select the model order from a range of candidate models. AIC, only successfully selects the correct model order by 42 times.

Next, to compare the influence of overlap to the three algorithms, a distribution fitted by two Gaussian components is generated with 200 samples. The two Gaussian components have the same mixture weights and unit diagonal covariance matrixes. The parameters of the two Gaussian components are:

$$\alpha_1 = 0.5, \alpha_2 = 0.5, u_1 = (0,0)^T, u_2 = (\delta, 0)^T$$

δ is a variable used to investigate its influence to the algorithm. The follow table shows the performance of three algorithm when the extent of overlap varies. It shows that the performance of AIC is influenced most by the overlap and BIC less. The degenerating model algorithm maintains a certain performance when the overlap increases.

Table 2. The influence of overlap to the algorithms

Performance	DM algorithm	BIC	AIC
$\delta = 2$	96%	86%	76%
$\delta = 1$	88%	70%	56%
$\delta = 0.5$	66%	54%	16%

Note: DM is the abbreviation of degenerating model

It could be seen from the table that with the extent of overlap increase, the performance of the three algorithms has some decline. When $\delta = 2$, of the 50 operation times, BIC fails 7 times, AIC fails 12 times and the degenerating model algorithm fails 2 times. When $\delta = 1$, BIC fails 15 times, AIC fails 22 times and the degenerating model algorithm fails 6 times. When $\delta = 0.5$, BIC fails 23 times, AIC fails 42 times and the degenerating model algorithm fails 17 times. The experimental result shows that comparing to AIC and BIC, the degenerating model algorithm has evident advantage to cope with overlap.

Experiment 3

As an application, the proposed algorithm is used for image segmentation in the third experiment. The purpose of image segmentation is to partition an image into meaningful regions with respect to a particular application and assign the pixels in an image to a relatively small number of groups where each group represents a set of pixels that are coherent in their gray-scale and other properties. Gaussian mixture model is used for image segmentation in this experiment. The procedure of image segmentation using GMM has been introduced in the former sections.

The main purpose of this experiment is to discuss whether the degenerating model algorithm could select the optimal model order, *i.e.*, the optimal number of segmentation region for a given image.

A 512*512 pepper image is used in the experiment.

The grey-level threshold of histogram-based technique is first used to determine the segmentation number for the given image.

The segmentation of the image is determined by the following criterion.

$$g(m,n) = \begin{cases} k, & T_{k-1} < f(m,n) \le T_k \\ 0, & f(m,n) \le T_0 \end{cases}$$

$T_0, T_1, T_2 \cdots T_k$ is a set of thresholds of the gray level. They are the peak valleys of the histogram and k is the label of the segmentation regions.

The histogram of the given image was drawn in Fig.3. The number of segmentation regions is selected by counting the "valley" between two peaks in the histogram.

From the histogram, it can be seen that the number of peak valleys of the origin image is roughly 3 or 4. Fig.4 and Fig.5 are the segmentation effect with segmentation number 3 and 4 respectively. Obviously, Fig.5 has better segmentation effect.

Then the degenerating model algorithm is used to calculate the segmentation number. As expected, the degenerating model algorithm run 10 times and every time it obtained the model order $M = 4$. Fig.6 is the segmentation effect using 4 mixture components GMM. This result shows the validity and robustness of the degenerating model algorithm.

AIC is then used to compute the number of segmentation number. The criterion of AIC is written as $-L(X, \Theta) + 2p$. It runs 10 times and the results are list in Table 3. The table shows the AIC has the tendency of over-fit the data. This is consistent with the experiments of model order selection using AIC that when the training data set is large, it often select a larger model order.

Table 3. Model order selection using AIC

Times	6	3	1
Model order	4	6	7

V. Conclusions

This paper proposes a new algorithm for unsupervised learning of Gaussian mixture model. It could simultaneously select the optimal model order and estimates the parameters of the Gaussian mixture model. The algorithm is based on the property of Kotz type distribution when its parameter $s = 1$, $r = 1/2$, $K \in [1, H]$. Such Kotz type distribution is named de-

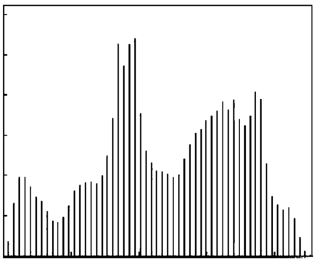

Fig. 3. Histogram of original image

Fig. 4. Segmentation with $k = 3$

Fig. 5. Segmentation with $k = 4$

Fig. 6. Segmentation using GMM ($M = 4$)

generating model when used to model order selection for GMM. When GMM is under saturated, the degenerating model would degenerate into a Gaussian component. When GMM is over-saturated, the degenerating model would show $K > 1$ and the training vectors would fall the degenerating model's crater region. Unlike other algorithms for model order selection and parameters estimation, it could simultaneously estimate the parameters of GMM and select the optimal model order. Experiment shows the algorithm achieves good performance in image segmentation.

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